

Master Thesis

Robust Attention State Prediction from EEG Signals Using Meta-Learning and Uncertainty-Aware Deep Learning

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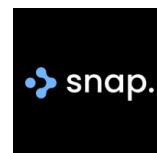
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Abstract

Electroencephalography (EEG)-based attention monitoring systems are increasingly relevant for safety-critical applications such as driver monitoring and human-machine interaction. However, current brain-computer interface (BCI) approaches face three fundamental challenges: limited generalization due to single-paradigm training, model training using predefined attention labels, and the absence of predictive uncertainty quantification for prediction models.

This thesis proposes a two-stage, uncertainty-aware meta-learning framework for robust attention state prediction from multi-paradigm EEG recordings. EEG data was collected from nine subjects across multi-paradigm video recording. In the first stage, attention states were discovered in a data-driven manner using unsupervised clustering on ground-truth spectral features (frequency bands), avoiding predefined labels. This process revealed three neurophysiologically distinct attention states: Baseline Attention, Low Arousal, and High Attention.

In the second stage, a spectral specialist based on Light Gradient Boosting Machine (LightGBM) was trained to predict band powers and ratios from EEG features. These predicted spectral features were then combined with contextual information and used by three meta-learners: Logistic Regression, XGBoost, and a Transformer, to classify attention states. Leave-One-Subject-Out cross-validation demonstrated strong cross-subject and cross-paradigm generalization, with Logistic Regression achieving the highest classification accuracy of 97.4%.

To enable safe and interpretable deployment, predictive uncertainty was explicitly quantified using model-specific methods, including calibrated entropy, ensemble-based uncertainty, and monte carlo dropout. Across all models, incorrect predictions exhibited 2–4 times higher uncertainty than correct predictions (with a significant p -value), indicating reliable uncertainty separation.

Overall, this work demonstrates that combining data-driven attention state discovery, cross-paradigm generalization, and systematic uncertainty quantification enables robust and interpretable EEG-based attention monitoring, advancing the practical applicability of BCI systems in real-world and safety-critical environments.

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Glossary

Term	Definition
BCI	Brain-Computer Interface
CHI	Calinski-Harabasz Index, clustering quality metric
EEG	Electroencephalography, non-invasive brain activity recording
ERP	Event-Related Potential, time-locked brain response to a stimulus
FFN	Feedforward Neural Network
HCI	Human-Computer Interaction
LightGBM	Light Gradient Boosting Machine, A gradient-boosting framework for machine learning
LOSO	Leave-One-Subject-Out cross-validation
MAE	Mean Absolute Error, regression metric
MC-Dropout	Monte Carlo Dropout, uncertainty quantification via stochastic inference
Meta-learner	Model combining outputs from multiple specialists for classification
P300	ERP component observed 300ms post-stimulus, attention marker
PSD	Power Spectral Density, frequency-domain signal representation
R^2	Coefficient of determination, proportion of variance explained
ROI	Region of Interest, spatially grouped EEG channels
Spectral Specialist	LightGBM regressor predicting band powers from EEG features
Transformer	Attention-based neural network architecture
XGBoost	Extreme Gradient Boosting, tree-based ensemble method

1 Introduction

Human attention is often perceived in an intuitive and context-dependent manner, encompassing a wide range of everyday interpretations. The literature of psychology and neuroscience defines attention as a dynamic cognitive process that continuously fluctuates during task execution, regardless of task difficulty [21]. This definition implies that even when an individual is focused on a single task, their attention states shift periodically among the primary task, internal thoughts (mind-wandering), and external distractions.

Recent technological advances in areas such as Brain–Computer Interfaces (BCI) [18], Human–Computer Interaction (HCI) [23], and neuroergonomic monitoring [73] have enabled increasingly accurate assessment of individual cognitive and attentional states. Among these approaches, Electroencephalography (EEG) stands out for its non-invasive nature and ability to capture neural signatures associated with attention states. However, building robust attention-state prediction systems from EEG signals faces a fundamental challenge: attention manifests differently across different paradigms [34].

Since the interpretation of neural signatures depends on the experimental design and contextual conditions [37], traditional machine learning and deep learning models are typically trained within a single paradigm to maintain uniform conditions. This thesis addresses the cross-paradigm generalization challenge by proposing a meta-learning framework for multi-paradigm EEG recordings with integrated model-specific uncertainty quantification. The proposed framework first discovers neurophysiologically grounded attention states through unsupervised clustering of EEG band-power features. It then learns to classify these states using predicted spectral features and contextual information, while explicitly quantifying model-specific prediction uncertainty.

Following this introductory chapter, the remainder of the thesis is structured as follows: Chapter 2 provides a comprehensive overview of the literature on EEG-based attention modeling, discussing the various techniques used in prior studies for attention modeling

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along with the gaps in those studies. Chapter 3 describes the mathematical foundations for algorithms used for attention classifications alongside evaluation metrics used to assess their quality. Chapter 4 outlines the experimental design, including data acquisition, preprocessing, feature extraction, attention state discovery, architecture of the classification framework, and cross-validation strategy. Chapter 5 presents results of each model and evaluates the performance of each model. Chapter 6 gives an overview of the thesis's findings and suggests future work.

2 Literature Review

2.1 EEG-Based Brain-Computer Interfaces:

The brain-computer interface (BCI), also known as brain-machine interface (BMI), is a system that translates the signals generated by the brain while performing an activity. EEG records this activity by measuring electrical signals on the scalp using electrodes. The BCI system interprets EEG recordings and uses them to control external devices or monitor cognitive states without involving any muscle or body movement [71]. Over the last few decades, the applications of BCI have evolved from performing lab-designed tasks to real-world applications, including communication devices for patients with paralysis [12], rehabilitation tools for stroke recovery [16], and a cognitive state monitoring system [80, 53].

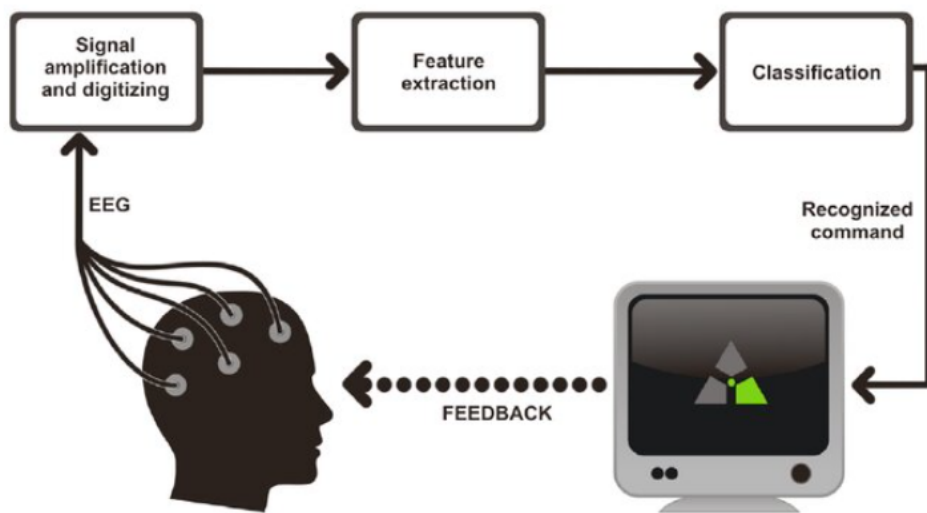


Figure 2.1: General scheme of an EEG-based BCI [13]

BCI systems are crucial in the safety-critical domain, where human inattention could lead to fatal outcomes. Driver attention monitoring systems use an EEG-based BCI to detect when a person is distracted or about to fall asleep, potentially preventing accidents [44, 51]. Similarly, the pilot workload monitoring system can detect when a pilot's elevated mental workload may lead to an error and trigger assistance systems to avoid any fatal outcomes [45]. In medical centers, the EEG-based sedation level tracking systems help in optimizing patients' drug dosage [82]. All these aforementioned applications share a common mandatory prerequisite that the system should have a high predictive accuracy, along with the knowledge of how certain it is about the prediction. Despite their potential, BCI systems face significant challenges in real-world deployment. This is primarily because EEG signals are inherently noisy, containing artifacts from sudden drifts, eye blinks, muscle activity, and electrical interferences [4]. Neural signatures from the brain vary across experimental subjects due to scalp conductivity, requiring time-consuming calibrations for each new subject [7]. Most critically, models that are trained on a single attention-related task fail in real-world scenarios where there is a sudden shift in attention states due to multiple stimuli [5].

2.2 Neural Signatures and their association with Attention States:

The human brain generates repetitive patterns of neural activity or brainwaves, which can be decomposed into frequency bands called brain oscillations or rhythms. These oscillations are associated with distinct cognitive and attentional states of a person [42]. These brain rhythms are commonly decomposed into five major frequency bands: delta (0.5–4 Hz), theta (4–7 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (30–40 Hz). These frequency bands have well-documented associations with arousal, attention, and cognitive effort. Alpha rhythms typically increase during relaxed wakefulness and decrease during externally directed attention, whereas beta activity is often enhanced during focused processing and high attention. The theta band is associated with relaxed and early stages of sleep. An increase in theta rhythms indicates a relaxed or mind-wandering state. High delta oscillations are associated with sleep and drowsiness, and Gamma oscillations are an indicator of high-level cognitive processing [8, 68, 2, 31].

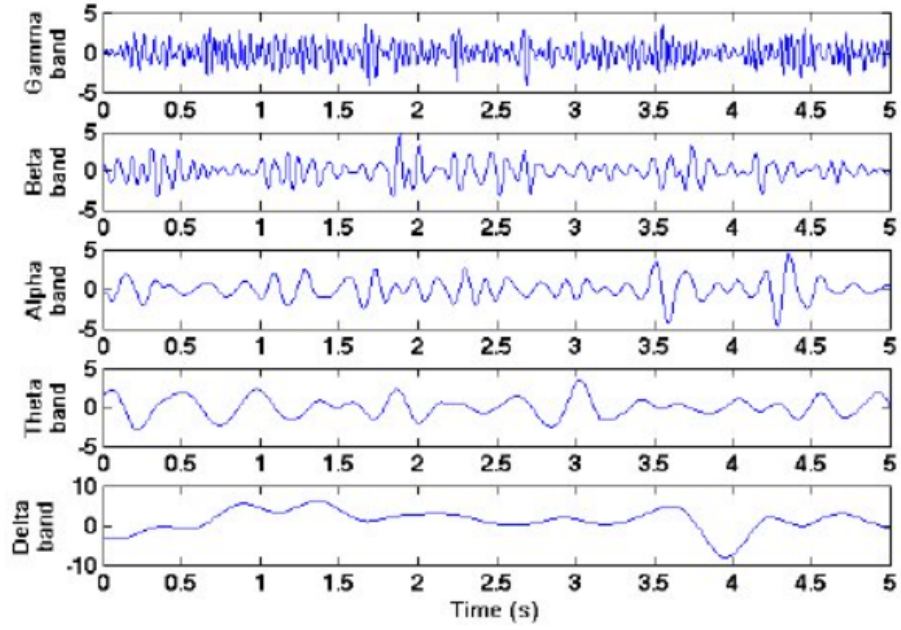


Figure 2.2: A visual depiction of five frequency bands for an EEG signal [3]

Along with the individual band values, the band ratios are also an indicator of attention states. Relaxed or drowsy states typically show high (α/β) ratios and low beta power, while attentive states will follow the opposite pattern [9]. Similarly, the theta-beta ratio θ/β , which helps in detecting drowsiness, has a high value when a person is fatigued [33].

2.3 Paradigms for Attention Detection:

In EEG, different experimental tasks are used for attention modeling. The most common is the oddball task, where a target occurs less frequently than a non-target, for example, pressing a button after seeing a letter 'A' (20% of trials) on the screen but not 'B' (80% of trials). When the subject presses the button, a brain response signal called P300 is produced, which is observed 300ms after each stimulus, indicating that the subject noticed it [48].

Oddball tasks have limited ecological validity in real-world scenarios, whereas the naturalistic viewing tasks provide a more realistic experimental condition. In these tasks, participants watch videos containing multiple unexpected events while EEG signals are recorded. This approach effectively captures the variability in attentional states elicited

by unexpected stimuli, mirroring responses observed in real-life settings [75].

For attention modeling, most existing studies are based on a single experimental paradigm, as combining multiple paradigms substantially complicates analysis [46]. However, this is the opposite of real-world scenarios such as a driver’s attention monitoring task, where individuals are exposed to multiple contexts simultaneously. As a result, models trained on a single task often fail when deployed in such settings [22].

2.4 Machine learning and Deep learning for EEG:

For EEG classification tasks, traditional machine learning methods such as linear discriminant analysis (LDA), support vector machines (SVM), and random forests have been widely used. These methods are typically applied after features have been extracted from EEG signals and are effective at learning patterns from these preprocessed inputs [58, 28]. Gradient boosting methods like XGBoost and LightGBM are often preferred when working with structured data, such as band-power values extracted from EEG signals. These boosting methods combine multiple decision trees to improve prediction accuracy [43, 35, 81]. Deep learning models, including recurrent neural networks (RNNs), convolutional neural networks (CNNs), and transformers, are capable of learning discriminative patterns directly from raw EEG data, often providing an advantage over traditional machine learning methods in BCI applications. However, when there is not enough training data, these models may fail to learn meaningful patterns and can perform worse than simpler machine learning methods that require less data. [67, 63, 66]. Meta-learning works by combining the outputs of multiple models rather than depending on just one. Since each model learns different aspects of the data, meta-learning learns how to weight and combine these predictions, leading to more reliable and accurate results [79, 30].

2.5 Uncertainty Quantification in Machine and Deep learning:

Uncertainty in machine learning models is commonly categorized into aleatoric and epistemic uncertainty. Aleatoric (data) uncertainty represents the irreducible noise inherent in the data. In contrast, epistemic or knowledge-based uncertainty arises from limited

knowledge about the problem at hand, such as insufficient training data, incomplete model understanding, or inaccuracies introduced during the modeling process [1, 69].

To optimize model performance, the uncertainty associated with model predictions is quantified using various techniques. In probabilistic models, uncertainty can be estimated using probability calibration methods, such as Platt scaling, which align predicted probabilities with observed outcomes [55]. In ensemble models, uncertainty is quantified by measuring the level of disagreement among the predictions of multiple models; stronger disagreement indicates higher uncertainty [61, 78]. For neural networks, Monte-Carlo dropout estimates uncertainty by randomly deactivating neurons during inference, where the resulting variability in predictions is interpreted as the model’s uncertainty [27].

2.6 Research Gaps and Thesis Motivation:

Although significant advances have been made in EEG-based attention modeling, critical challenges remain when deploying robust attention monitoring systems in real-world settings. First, most studies focus on a single paradigm in isolation to simplify analysis. However, this is insufficient for real-world applications, where attention continuously shifts across different contexts and tasks. Second, many studies rely on predefined attention labels, such as assigning “high attention” to target trials. This causes models to focus on predicting labels rather than discovering the underlying cognitive states and patterns present in the EEG data. Lastly, most studies focus primarily on improving prediction accuracy while largely neglecting the uncertainty associated with those predictions. This is a critical limitation for real-world, safety-critical applications, where confident but incorrect predictions made without reliable uncertainty estimates could lead to serious or even fatal consequences.

This thesis addressed these gaps through a meta-learning framework that integrates spectral predictions with contextual information to classify attention states of experimental subjects, along with model-specific uncertainty quantification. Instead of relying on predefined attention labels or fixed threshold-based definitions, we applied unsupervised clustering to ground-truth EEG band-power features to identify three neurophysiologically distinct attention states. These states were then subsequently ordered using the alpha–beta band power ratio. By allowing the neural patterns themselves to define the

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attention states, this approach avoided the circular reasoning that is inherent in predefined labeling schemes.

For achieving cross-paradigm generalization, we combined the oddball task and naturalistic video task into a unified experimental framework. We then trained models across these multiple contexts and validated their generalization capabilities using the LOSO cross-validation strategy. Lastly, we used model-specific uncertainty quantification methods to estimate predictive uncertainty associated with each model’s predictions.

3 Methods Used

3.1 Clustering:

Clustering, also known as data segmentation, is an unsupervised learning technique that combines groups of data points with similar characteristics into clusters. The goal of clustering is to minimize the distance between data points assigned to the same cluster and maximize the distance between clusters, thereby achieving a well-defined cluster separation [32].

For this thesis, we used one of the most well-known clustering algorithm, k-means clustering, which is well-suited for working with quantitative data such as EEG.

3.1.1 k-means Clustering:

In k-means clustering, the algorithm initially creates k clusters. It then initializes μ_k centroid. Then, it iterates in two steps: first, it assigns each feature vector to its nearest centroid, and second, it recomputes each centroid as the mean of all assigned vectors belonging to that cluster [32].

Let $\{\mathbf{f}_i\}_{i=1}^N$ represent a feature vector for epoch i (section 4.4.1). The k-means clustering partitions these features into k clusters by minimizing the intra-cluster variance.

$$\min_{\{\mu_k\}_{k=1}^K} \sum_{i=1}^N \|\mathbf{w}(\mathbf{f}_i - \mu_{c(i)})\|^2 \quad (3.1)$$

where μ_k represents centroid of cluster k , $\mathbf{w}$ is weight vector for features, i represents each individual epoch, and $c(i)$ is the cluster assignment of epoch i .

In this thesis, the feature vector f_i corresponds to a seven-dimensional feature target vector y_{true} as defined by the equation 4.12, comprising five relative band powers and two attention-relevant ratios.

3.1.2 Validation Metrics for Clustering:

Silhouette Coefficient:

Proposed in 1987 by Rousseeuw [62], the silhouette coefficient is a metric to define clustering quality. It does so by first quantifying how closely the data points within a single cluster are grouped (cohesion) and how well separated each cluster is from the others (separation).

Let $a(i)$ be the mean intra-cluster distance for a sample i_k in cluster k and $b(i)$ be the average distance for sample i_k from all samples in the nearest cluster j , then the silhouette coefficient is calculated as

$$S(i) = \frac{b(i) - a(i)}{\max a(i), b(i)} \quad (3.2)$$

The value of the silhouette coefficient ranges from -1 to +1, where +1 indicates excellent clustering, and -1 points towards misclassification. Values near zero indicate overlapping clusters.

Calinski–Harabasz Index:

The Calinski–Harabasz index is another metric for clustering evaluation. Introduced by Calinski and Harabasz in 1974 [15], this metric is calculated as the ratio of the sum of inter-cluster dispersion and the sum of intra-cluster dispersion for all clusters.

For N samples distributed into k clusters, the Calinski–Harabasz index is given as

$$CH = \frac{SS_B/(k-1)}{SS_W/(N-k)} \quad (3.3)$$

where SS_B is the sum of squares between clusters and SS_W is the sum of squares within a cluster.

3.2 Gradient Boosting:

An ensemble learning method that takes multiple weak learners (usually tree-based learners) and combines them to construct a strong predictive model. The gradient boosting builds models sequentially, where each new model corrects the prediction errors made by the previous ensemble [52].

3.2.1 Light Gradient Boosting Machine (LightGBM):

For a training dataset $\{(x_i, y_i)\}_{i=1}^N$ where $x_i \in \mathbb{R}^d$ is the input feature vector and y_i is the target prediction, lightGBM constructs an ensemble model

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i) \quad (3.4)$$

where K is the total number of trees and f_k represents k -th tree in the ensemble. As a leaf-wise gradient boosting framework, LightGBM enables fast and memory-efficient learning while achieving strong performance on high-dimensional EEG data [38].

During each training iteration, the model minimizes the loss function $L(y, \hat{y})$. The loss function measuring the prediction error is given as:

$$\mathcal{L}^{(k)} = \sum_{i=1}^N L(y_i, \hat{y}_i^{(k-1)} + f_k(x_i)) + \Omega(f_k) \quad (3.5)$$

where N is the number of training samples (epochs in this thesis), y_i is the true target value for sample i , $\hat{y}_i^{(k-1)}$ is the prediction before adding k -th tree, $f_k(x_i)$ is the new tree added during k -th iteration and $\Omega(f_k)$ is the regularization term that controls model's complexity.

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (3.6)$$

where T is the number of leaves in tree f , w_j is the weight assigned to the leaf, and γ, λ are regularization hyperparameters.

3.2.2 XGBoost Classifier:

A classification technique that builds an ensemble of decision trees sequentially, where each new tree is trained to correct the errors of the existing ensemble [17]. For a multiclass classification task, let K be the total number of classes, $\mathbf{x}$ be the input feature vector. The XGBoost classifier will then generate a vector of K class membership scores

$$\mathbf{s}(\mathbf{x}) = [s_1(\mathbf{x}), s_2(\mathbf{x}), \dots, s_K(\mathbf{x})] \quad (3.7)$$

where each score $s_k(\mathbf{x})$ represents the model's confidence for class K . These scores are

3 Methods Used

calculated by summing the outputs of multiple trees belonging to that class.

$$s_k(\mathbf{x}) = \sum_{i=1}^T f_{k,i}(\mathbf{x}) \quad (3.8)$$

where T is the number of boosting iteration and $f_{k,i}$ represents the i -th tree being trained for class k .

To transform these output scores into probabilities, the XGBoost classifier uses the softmax function, ensuring that these output scores form a valid probability distribution over classes.

$$P(y = k \mid \mathbf{x}) = \frac{\exp(s_k(\mathbf{x}))}{\sum_{j=1}^K \exp(s_j(\mathbf{x}))} \quad (3.9)$$

During training, the classifier tries to minimize the multiclass log-loss with regularization.

$$\mathcal{L} = - \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log(P(y = k \mid \mathbf{x}_i)) + \sum_{t=1}^T \Omega(f_t) \quad (3.10)$$

where y_{ik} is equal to 1 if sample i belongs to class k and 0 otherwise. The term $\Omega(f_t)$ is a regularization term that controls the classifier's complexity by penalizing large trees and extreme leaf weights, thereby improving the model's prediction capabilities.

3.3 Transformer:

A transformer is a neural network built around a self-attention mechanism. In contrast to vanilla RNNs and LSTMs, which operate sequentially, a transformer processes inputs in parallel, resulting in more stable training [70].

Given an input sequence vector $\mathbf{x} \in \mathbb{R}^d$ (which in this thesis is a feature vector of length one), the self-attention calculates the weighted representation by measuring relevance between different parts of the input. For this purpose, it first projects the input into three representations:

$$\mathbf{Q} = \mathbf{x}W^Q, \quad \mathbf{K} = \mathbf{x}W^K, \quad \mathbf{V} = \mathbf{x}W^V, \quad (3.11)$$

where $\mathbf{Q}$ is the query vector which describes the information we are looking for. $\mathbf{K}$ is the key vector describing the information that we have, and $\mathbf{V}$ is the value vector which

3 Methods Used

represents the content that is passed forward. The attention weights between elements are then computed using

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (3.12)$$

where d_k is the dimension of the key vector. The scaling factor $\frac{1}{\sqrt{d_k}}$ prevents the dot products from growing too large, which would otherwise hinder the model's training.

Rather than a single attention operation, the transformer employs a multi-head attention, which allows the model to attend to different features simultaneously. For each attention head h , separate projections are learned:

$$\text{head}_h = \text{Attention}(QW_h^Q, KW_h^K, VW_h^V). \quad (3.13)$$

All outputs are then concatenated and linearly transformed via:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W^O \quad (3.14)$$

where

$$W^O \in \mathbb{R}$$

projects the combined representation back to original feature dimension.

Next, we have an encoder layer that comprises two sublayers. In the first sublayer, multihead self-attention is applied:

$$\mathbf{z} = \mathbf{x} + \text{MultiHead}(\mathbf{x}, \mathbf{x}, \mathbf{x}), \quad (3.15)$$

followed by a position-wise feedforward neural network:

$$\text{output} = \mathbf{z} + \text{FFN}(\text{LayerNorm}(\mathbf{z})). \quad (3.16)$$

The feedforward neural network is usually defined as:

$$\text{FFN}(\mathbf{z}) = \text{ReLU}(\mathbf{z}W_1 + \mathbf{b}_1)W_2 + \mathbf{b}_2, \quad (3.17)$$

where $W_1 \in \mathbb{R}$ expands representation to higher dimension and $W_2 \in \mathbb{R}^{d_{ff} \times d}$ projects it back to original dimension. After the encoder layer, a layer normalization function is performed for stabilizing the training loop and to improve the model's convergence:

$$\text{LayerNorm}(\mathbf{x}) = \gamma \times \frac{\mathbf{x} - \mu}{\sigma} + \beta, \quad (3.18)$$

where μ is the mean and σ is the standard deviation calculated across feature dimensions, and β and γ are scaling and shifting parameters.

For predicting the K classes, the output of the final encoder layer is first passed through a linear function, which outputs the class scores (logits). These logits are then converted into class probabilities using the softmax function:

$$P(y = k \mid \mathbf{x}) = \text{softmax}(W_{\text{class}}\mathbf{z} + \mathbf{b}_{\text{class}}), \quad (3.19)$$

where $\mathbf{z}$ denotes the output representation of encoder layer, W_{class} maps this representation to K class logits, and $\mathbf{b}_{\text{class}}$ is the bias term.

3.4 Evaluation Metrics:

3.4.1 Statistical Metrics:

Shapiro-Wilk-Test for Normality:

The Shapiro-Wilk test examines whether a sample follows a normal distribution. The null hypothesis for the Shapiro-Wilk test is that the sample belongs to a normal distribution. As a rule of thumb, we reject the null hypothesis if $p < 0.05$.

For the Shapiro test, we calculate a W statistic that tests the condition of normality of the sample $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$. Small values of W indicate deviation from normality. The W statistic is calculated as follows:

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (3.20)$$

where the $x_{(i)}$ is the i -th smallest value in the given sample, $\bar{x}$ is the sample mean, and the a_i are constants generated from the means, variances, and covariances of the order statistics of a sample of size n from a normal distribution [26].

Levene's Test:

Levene's test is a parametric test used to check if k samples have equal variances. Equal variances across samples are called homogeneity of variance or homoscedasticity. Some

3 Methods Used

statistical tests, for example, the analysis of variance (ANOVA), assume that variances are equal across groups or samples. The Levene's test can be used to verify that assumption [26]. The null hypothesis for the Levene test is defined as:

$$\begin{aligned} H_0 : \quad & \sigma_1^2 = \sigma_2^2 = \dots = \sigma_k^2 \\ H_A : \quad & \sigma_i^2 \neq \sigma_j^2 \quad \text{for at least one pair } (i, j) \end{aligned}$$

where σ_i is variance of population for i -th group and k represents total number of groups. For test statistics, we first calculate group-centered absolute deviations for every observation j in group i using:

$$Z_{ij} = |X_{ij} - \bar{X}_i|$$

where X_{ij} is j -th observation in i -th group. and $\bar{X}$ is group mean. Next, we calculate the test statistic for Levene's test as:

$$W = \frac{\sum_{i=1}^k n_i (\bar{Z}_i - \bar{Z})^2 / (k - 1)}{\sum_{i=1}^k \sum_{j=1}^{n_i} (Z_{ij} - \bar{Z}_i)^2 / (N - k)} \quad (3.21)$$

where $\bar{Z}_i$ is the mean deviations in group i , $\bar{Z}$ is mean deviation of all k groups, n_i is number of observations in i -th group and N is the total sample size [83].

Independent t-test:

The independent t-test, also known as the two-sample t-test or Student's t-test, is an inferential statistical test that determines the difference in means of two independent groups [39].

The null hypothesis for the t-test states that there is no difference in population means, and the alternative hypothesis states that population means are not equal. The null hypothesis and alternative hypothesis for the independent sample t-test are given as:

$$\begin{aligned} \text{Null Hypothesis: } & H_0 : \mu_1 = \mu_2 \\ \text{Alternative Hypothesis: } & H_1 : \mu_1 \neq \mu_2 \quad (\text{two-tailed}) \end{aligned}$$

The test statistics for an independent two-sample t-test with equal variance are given

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as:

$$t = \frac{\bar{x} - \bar{y}}{SE} \quad (3.22)$$

where $\bar{x}$ and $\bar{y}$ are means of the two samples, and SE represents the standard error of the mean difference, which is derived from the pooled variance s_p^2 , which is calculated as:

$$s_p^2 = \frac{(n_1 - 1)s_x^2 + (n_2 - 1)s_y^2}{df} \quad (3.23)$$

where s_x^2 and s_y^2 are the sample variances and df is the degree of freedom. After calculating the pooled variance s_p^2 , we then calculate the standard error of the mean difference using:

$$SE = \sqrt{s_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} \quad (3.24)$$

where n_1 and n_2 are sample sizes. The degree of freedom df is calculated as:

$$df = n_1 + n_2 - 2 \quad (3.25)$$

Once the t-statistics and degrees of freedom are calculated, we determine the two-sided p -value using:

$$p = 2 \cdot P(T_{df} \geq |t|) \quad (3.26)$$

We reject the null hypothesis H_0 if the p value is less than the significance level α .

Cohen's d for Effect Size:

Cohen's d is a measure of effect size, which determines how large the difference is between the means of two groups based on standard deviation [24]. Mathematically, it is calculated as:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p} \quad (3.27)$$

where $\bar{x}_1$ and $\bar{x}_2$ are the means of two groups and s_p is the pooled standard deviation. The value of Cohen's d can be interpreted categorically as follows: a value of $d = 0.2$ represents a small effect, $d = 0.5$ a medium effect, and $d = 0.8$ indicates a large effect.

Analysis of Variance (ANOVA):

The one-way analysis of variance test, or ANOVA, is a statistical measure to compare the means of k populations where $k \geq 3$ [40]. The assumptions of the test are as follows:

- The observations are random, and the samples observed from the k populations are independent of each other.
- The data should be normally distributed, and follow the condition of homoscedasticity.

The hypotheses are:

Null Hypothesis: $H_0 : \mu_1 = \mu_2 = \dots = \mu_k$

Alternative Hypothesis: $H_1 : \text{At least one pair } (\mu_i, \mu_j) \text{ is unequal, where } i \neq j \text{ and } i, j \in \{1, \dots, k\}.$

The one-way ANOVA is calculated using the F-statistic. It is given as:

$$F = \frac{\sum_{j=1}^k n_j (\bar{x}_j - \bar{\bar{x}})^2 / (k - 1)}{\sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2 / (N - k)} \quad (3.28)$$

where n_j is the sample size of the j -th group; $\bar{x}_j$ is the sample mean of the j -th group; $\bar{\bar{x}}$ is the overall mean for all groups; k is the number of groups; x_{ij} is the i -th observation of the j -th group; and N is the number of observations in all groups.

3.4.2 Regression Metrics:

Regression metrics are an indicator of how well a model predicts the continuous target values $\hat{y}$. In this thesis, we used the following regression metrics.

Mean Absolute Error:

Mean absolute error, or MAE (also known as L1 loss), is a measure of the average magnitude of prediction errors without including their direction [77]. We use MAE because it minimizes the influence of large errors and is robust to outliers. It is computed as:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (3.29)$$

where y_i is the true value and $\hat{y}_i$ is the predicted value and N is the number of input samples.

Coefficient of Determination (R^2):

The R^2 score measures how much a model can capture variance in the underlying data. It quantifies the proportion of variance in the data that is explained by the model, providing a measure of how well a model's predictions align with true target values [25]. An R^2 value of 1 indicates a perfect prediction, whereas a value of 0 indicates that the model's performance is no better than simply predicting the mean of the target variable. Mathematically, R^2 is calculated as:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}} \quad (3.30)$$

where SS_{res} is the residual sum of squares, calculated as:

$$SS_{res} = \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (3.31)$$

and SS_{tot} is the total sum of squares calculated as:

$$SS_{tot} = \sum_{i=1}^N (y_i - \bar{y})^2 \quad (3.32)$$

the term y_i is the each individual observation, $\hat{y}_i$ is the predicted value and $\bar{y}$ is the mean of the true values.

3.4.3 Classification Metrics:

In classification tasks, the classification learners assign discrete class labels to input instances [65]. In this thesis, we used multiple classification metrics as indicators of how well our meta-models can predict the attention states of each individual.

Accuracy:

Accuracy quantifies the proportion of correctly classified instances as:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(y_i = \hat{y}_i) \quad (3.33)$$

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where $\mathbb{I}[\cdot]$ is an indicator function and outputs 1 when the learner predicts the correct class label and 0 otherwise.

Precision and Recall:

For each class k , precision describes, of all the instances that are predicted as class k , how many of them actually belong to class k . High precision indicates few false positives. We calculate precision using:

$$\text{Precision}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k} \quad (3.34)$$

whereas Recall measures, among all the true instances of class k , what fraction is accurately classified as class k . High recall indicates few false negatives. We determine recall as:

$$\text{Recall}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FN}_k} \quad (3.35)$$

the term TP_k is true positive for class k (an instance correctly predicted as k). FP_k is false positive for class k (an instance incorrectly predicted as class k). FN_k is a false negative for class k (an instance belongs to class k but classified as doesn't belong to class k).

F1-Score

It is a harmonic mean of precision and recall. We calculate the F1-score as:

$$F1_k = 2 \times \frac{\text{Precision}_k \times \text{Recall}_k}{\text{Precision}_k + \text{Recall}_k} \quad (3.36)$$

The motivation behind using the F1-score is that it provides a balance between recall and precision, capturing both false positives and false negatives

Macro-F1

To weigh each class equally, irrespective of its frequency, we use the macro-F1 score [10]. It is defined as:

$$F1_{\text{macro}} = \frac{1}{K} \sum_{k=1}^K F1_k \quad (3.37)$$

where $F1_k$ is F1 score for class k .

3.5 Uncertainty Quantification Methods:

3.5.1 Platt Scaling:

Platt scaling is a probability calibration method that converts a model's output score into a calibrated class probability using a sigmoid function to match the actual observed frequency of the predicted class [55]. Mathematically:

$$P(y = 1 | s) = \frac{1}{1 + \exp(As + B)} \quad (3.38)$$

where s is the raw model score and A, B are parameters learned from a held-out calibration set. For a multiclass classification task, calibration is applied to each one-vs-rest class independently.

3.5.2 Monte Carlo Dropout:

Monte Carlo dropout is a stochastic method for quantifying predictive uncertainty in deep learning models. The core idea of MC dropout is to activate dropout at test time and perform multiple forward passes through the network, which results in a distribution of predictions from which predictive uncertainty can be quantified [64].

For a neural network $f(\mathbf{x}; W)$ where W represents the model's weights, applying dropout during N forward passes will result in N stochastic prediction i.e. $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_N$. We then quantify uncertainty using the mean and variance across predictions:

$$\begin{aligned} \mathbb{E}[y|\mathbf{x}] &\approx \frac{1}{N} \sum_{i=1}^N \hat{y}_i \\ \text{Var}[y|\mathbf{x}] &\approx \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - \mathbb{E}[y|\mathbf{x}])^2 \end{aligned} \quad (3.39)$$

where $\mathbb{E}[y|\mathbf{x}]$ is the predictive mean and $\text{Var}[y|\mathbf{x}]$ is the predictive variance [50]. A high uncertainty value indicates epistemic uncertainty, which is the model's lack of knowledge about the correct prediction due to insufficient training data, and can be reduced by having more training data.

3.5.3 Mann–Whitney U test:

Mann-Whitney U test is a nonparametric statistical test. The null hypothesis of this test assumes that the two groups X and Y come from the same probability distribution, implying that there is no systematic difference between the two groups. The alternative hypothesis states that both groups belong to different probability distributions [54]. Mathematically, the U-statistic is calculated as:

$$U_x = R_x - \frac{n_x(n_x + 1)}{2} \quad \text{and} \quad U_y = R_y - \frac{n_y(n_y + 1)}{2}$$

with the relationship:

$$U_x + U_y = n_x n_y. \quad (3.40)$$

and the test statistic is defined as:

$$U = \min(U_x, U_y). \quad (3.41)$$

where n_x is sample size of group X and n_y is sample size of group Y . R_x is the sum of the ranks assigned to the group, and R_y is the sum of the ranks assigned to the second group. After calculating U statistics and determining the significance level (α), we can reject the null hypothesis if the p-value associated with the smallest U-value is less than our predetermined significance level (usually $\alpha = 0.05$).

4 Experimental Design and Methodology

4.1 Research Objectives:

The experimental design for this thesis has four core objectives:

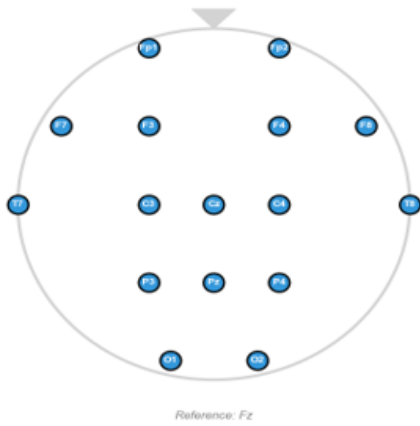
1. **Event-Related Potential (ERP) detection:** Confirming the presence of neural signatures, specifically the P300 response, which is a brain electrical response to external stimulus observed between 300-600ms post stimulus [6], via ERP visualization.
2. **Attention state discovery:** Verifying attention-related states present across subjects and visual paradigms by extracting EEG spectral profiles and their temporal dynamics.
3. **Meta-learning performance:** Assessing whether combining spectral features, specialist predictions, and additional contextual information improves classification accuracy of the derived attention states.
4. **Cross-subject generalization:** Evaluating whether the discovered states and the corresponding predictive models generalize unseen data/individuals well using a leave-one-subject-out (LOSO) validation strategy.

4.2 Data Acquisition:

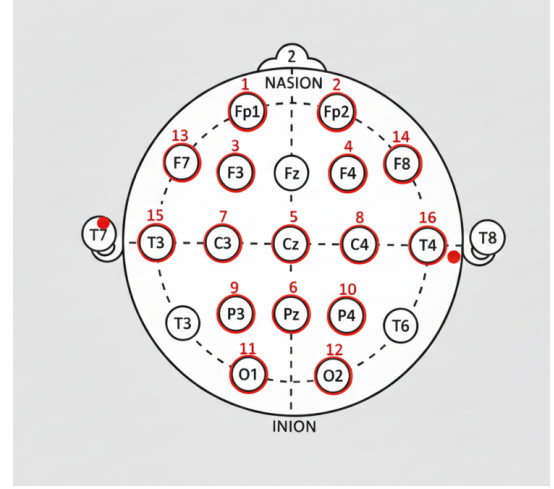
The dataset we used for this thesis is from in-house experimentation conducted at SNAP GmbH. The aim is to record the brain activity while subjects watch a continuous video containing multiple stimuli. Nine individuals (ages 20 to 40) participated in this study. All recordings were conducted in a quiet room to minimize external distractions and to reduce any additional noise in the EEG signals. All participants sat in front of a monitor at a fixed distance and were instructed to sit as still as possible.

4.2.1 EEG Hardware and Recording Setup:

We recorded EEG signals using an OpenBCI-based system equipped with sixteen active electrodes. We placed electrodes using the International 10-20 system, covering frontal, central, parietal, temporal, and occipital brain regions [41]. The electrode 'Fz' is set to be the reference electrode, and the sampling rate is set to 125 Hz. Each electrode reading is in microvolts (μV) and saved as a .csv file. Each CSV file contains raw electrode readings, sampling rate information, a time synchronization marker, and an event label indicating the current visual paradigm being used. A visual description of the electrode placement is provided in figure 4.1.



(a) Schematic view



(b) Labeled 10-20 system layout

Figure 4.1: EEG electrode placement according to the International 10-20 system. The 16 channels used in this study are highlighted in red.

4.2.2 Experimental Paradigms:

During a single continuous recording session, each participant watched the video clips from four stimulus categories to evoke distinct attention patterns:

- **Oddball paradigm:** A classical paradigm to evoke P300 response in an experimental subject[60]. In this paradigm, a rare target (red dot in this study) image appears among more frequent non-target (blue dot) images. All subjects are informed beforehand about the target and non-target stimuli.
- **Car crash paradigm:** A continuous, highly arousing visual sequence of car crash clips and normal driving clips, aimed to induce externally oriented attention in subjects [47].

- **Animal Surprise:** A naturalistic video segment containing a moving animal against scenes without the animal’s presence, designed to produce moderate attentional engagement.
- **Surprise Video:** A video segment features unexpected events such as a surprise appearance of a meteor and a bee in a static scenery to increase the subject’s attentional engagement.

Two out of nine participants are not able to complete all of the paradigms mentioned above due to a slight variation in the early recording sessions. Nevertheless, all the participants watched the oddball, car crash, and animal surprise clips.

4.3 EEG Preprocessing Pipeline:

Each raw EEG recording contains a mixture of neural activity and noise resulting from sudden muscle movement, eye blinks, and slow drifts. To ensure that noise artifacts present in the recordings do not interfere with data analysis and the inference of meaningful patterns, the data is passed through a structured preprocessing pipeline, which we created using EEG signal processing and neural time-series analysis literature [20, 76].

4.3.1 Data Import:

All raw EEG recordings are first loaded from the original CSV files, collected during data acquisition (section 4.2). Each EEG file had 16 channels and a sampling frequency of 125 Hz, a time index, and a column with discrete event labels. We then performed the following cleanup steps:

- **Numeric Retention** Only numeric EEG values are kept; all non-numeric values are discarded.
- **Missing Data** Rows containing missing values due to hardware interruptions are also dropped.
- **Sample Preservation:** All of the remaining samples are then kept to maintain the temporal continuity of each recording.

4.3.2 Timeline Creation:

An EEG signal consists of a sequence of voltage measurements, indexed by their position in the time series. To map these measurements to real time, we converted each index to its corresponding timestamp using the equation:

$$t = \frac{n}{f_s} \quad (4.1)$$

where t is the absolute time in seconds, n is the sample index, and f_s is the sampling rate. This temporal mapping synchronized each neural activity with experimental events throughout the recording session

4.3.3 Epoching:

After creating the timeline, we then divided each recording into multiple epochs. From EEG literature, an epoch is defined to be a short time-locked window around stimulus onset [48]. We then performed epoching on the entire EEG recordings, which enabled us to conduct event-related analyses and visualize ERPs.

4.3.4 Band-pass Filtering:

To eliminate slow DC drifts introduced by sudden movements and noise caused by high-frequency components during an EEG recording, we deployed a Butterworth band-pass filter. It limits the frequency between 0.5–45 Hz [14]. This defined range preserves the EEG frequency bands (section 2.2), which are the determinants of attentional states. To avoid phase distortions, we applied zero-phase forward and backward filtering [57]. This resulted in the removal of 96 μ V DC offsets from the raw EEG recordings.

4.3.5 Quality Assurance After Filtering:

After filtering, we performed a two-stage quality assurance check for each channel to identify and handle contaminated epochs. For this, we first segmented continuous EEG data into two-second windows with a step size of half a second. For each window, we calculated peak-to-peak(p2p) voltage as:

$$\text{p2p} = \max(x) - \min(x) \quad (4.2)$$

and standard deviation as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2} \quad (4.3)$$

where $x = \{x_1, x_2, \dots, x_N\}$ represents the filtered EEG signal within the analysis window, $\bar{x}$ is the mean, and N is the number of samples in the window. Quality control thresholds are set to $p2p < 500 \mu V$ and $\sigma < 200 \mu V$.

Windows that failed to meet these criteria are labeled as bad segments [56]. Next, these windows are merged with channel-specific bad segment masks, which pointed towards the temporal distribution of artifacts per channel. The fraction of contaminated windows then quantified channel quality as:

$$f_{\text{bad}} = \frac{1}{T} \sum_{t=1}^T \mathbb{I}[\text{window}_t \text{ is bad}] \quad (4.4)$$

where T is the total number of windows and $\mathbb{I}[\cdot]$ is the indicator function (1 if the condition is true, 0 otherwise). For channel retention, we followed established guidelines [11]:

- $f_{\text{bad}} < 0.30$: Retain channel with bad segment mask
- $0.30 \leq f_{\text{bad}} < 0.80$: Mark for interpolation
- $f_{\text{bad}} \geq 0.80$: Exclude channel from further analysis

Across all of nine subjects, all channels had $f_{\text{bad}} < 0.30\%$ and were included for further processing

4.3.6 Event Synchronization:

As we have multiple paradigms (section 4.2.2) from the thesis experimentation, we first reconstructed event timelines by extracting paradigm-specific labels from each recording. Then, we segmented the recording into paradigm-specific chunks: the Oddball task and video task (including car crashes, animal presence, surprise events). Minor variations, such as exclusion of surprise clips (section 4.2.2) in early recordings, are also taken into account. These alignments then allowed us to do uniform comparisons of EEG patterns across each paradigm.

4.3.7 Normalization:

Next, we performed data normalization to reduce inter-subject differences in each subject’s absolute amplitude and to improve consistency across each paradigm [20]. All channels are standardized using z-score normalization:

$$z = \frac{x - \mu}{\sigma} \quad (4.5)$$

where x represents the filtered EEG signal, μ is the channel-wise mean, and σ is the channel-wise standard deviation, both computed across the entire recording session for each subject independently.

4.4 Feature Extraction:

After the preprocessing pipeline, we transformed continuous EEG signals into composite features, which are then used for clustering and classification. The goal is to capture spectral characteristics of the signal alongside their attentional signatures [31, 20, 47]. These features are then used as inputs for attentional state discovery and meta models, explained later in this thesis.

4.4.1 Epoch Segmentation:

Since we recorded EEG continuously, each session’s recording is represented as a two-dimensional matrix with dimensions time(samples) and electrodes(channels). To perform analysis on these continuous recordings, we divided each recording into discrete epochs of fixed length. We set each epoch to be a one-second window without any overlaps. Since the sampling rate is set to 125 Hz, each epoch hence contained 125 samples per channel. For each epoch, we assigned an event label in accordance with the paradigm that is activated during that one-second interval. After epoch creation, each recording is converted to a three-dimensional matrix with axes corresponding to electrodes(channels), time(samples), and trials(epochs) [20].

4.4.2 Power Spectral Density Estimation:

Next, we estimated the power spectral density (PSD) using Welch’s method [74] to quantify the distribution of signal power across frequency bands. Welch’s method reduces spectral variance by dividing each epoch into overlapping segments, computing

the periodogram for each segment, and averaging the results [20].

For each epoch, the signal is divided into K overlapping segments of length one (125 samples at 125 Hz) with 50% overlap. The periodogram for each segment k is computed as:

$$P_k(f) = \frac{1}{L} \left| \sum_{n=0}^{L-1} w[n] \cdot x_k[n] \cdot e^{-j2\pi f n / f_s} \right|^2 \quad (4.6)$$

where $x_k[n]$ is the filtered and normalized EEG signal in segment k , $w[n]$ is a Hanning window, and f_s is the sampling rate. The final PSD estimate is obtained by averaging across segments:

$$\text{PSD}(f) = \frac{1}{K} \sum_{k=1}^K P_k(f) \quad (4.7)$$

We then integrated the PSD within each frequency band (section 2.2) to obtain band-specific power values.

4.4.3 Global and Relative Band Power Features:

Instead of analyzing the calculated PSD for each channel independently, we summed the band-specific power values across all sixteen channels to obtain a single global measure for each of the five frequency bands per epoch. This approach is motivated by two factors. First, attention-related activity tends to occur across distributed networks rather than isolated regions. Second, summing across channels enhances signal detection while minimizing random noise [20]. Additionally, this spatial summation reduced feature space from 80 values (16 channels $\times$ 5 bands) to just five, simplifying the analysis.

For each frequency band b , the global power is computed as:

$$P_b = \sum_{c=1}^{16} P_{b,c} \quad (4.8)$$

where $P_{b,c}$ is the power in band b for channel c . This yielded a five-dimensional global feature vector per epoch:

$$\mathbf{P}_{\text{global}} = [P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma] \quad (4.9)$$

representing the total power in each of the five frequency bands. These global band power values served two main purposes in this thesis: first, they acted as an intermediate step from which we derived relative powers and ratios relevant for predicting the attention states.

One challenge with the global band power values is their high inter-subject variability, which is attributed to factors such as skull conductivity, electrode impedance, and individual head morphology. To address these inter-subject dissimilarities, we computed relative power for each band using:

$$P_{b,\text{rel}} = \frac{P_b}{\sum_k P_k} \quad (4.10)$$

where $P_{b,\text{rel}}$ is the relative power of band b , P_b is the absolute power in that band, and $\sum_k P_k$ is the total power across all five frequency bands. This relative power measure enabled us to quantify the distribution of spectral energy across frequency bands. For example, a relative alpha power of 0.35 would indicate that 35% of the total brain activity for that person is alpha oscillations.

Alongside band power values, we measured alpha-beta and theta-beta ratios, which are established attention signatures as defined by EEG neuroscience literature [9, 19]

$$R_{\alpha/\beta} = \frac{P_{\alpha,\text{rel}}}{P_{\beta,\text{rel}}}, \quad R_{\theta/\beta} = \frac{P_{\theta,\text{rel}}}{P_{\beta,\text{rel}}} \quad (4.11)$$

where $P_{\alpha,\text{rel}}$, $P_{\theta,\text{rel}}$, and $P_{\beta,\text{rel}}$ are the relative powers in the alpha, theta, and beta bands, respectively. Both relative band power values and their derived ratios then formed a seven-dimensional target vector:

$$\mathbf{y}_{\text{true}} = [P_{\delta,\text{rel}}, P_{\theta,\text{rel}}, P_{\alpha,\text{rel}}, P_{\beta,\text{rel}}, P_{\gamma,\text{rel}}, R_{\alpha/\beta}, R_{\theta/\beta}] \quad (4.12)$$

The target vector $\mathbf{y}_{\text{true}}$ consists of five relative band powers and two relative band ratios, as these relative band powers are robust to inter-subject variability and provide a normalized representation of the spectral energy distribution. This seven-dimensional vector then serves two purposes: one, it acts as input for clustering for discovering the attention state labels, and second, it behaves as a target for the spectral specialist.

4.4.4 Region-of-Interest Band Power Features:

The neural attention signatures are generally associated with the spatial location of the channels on the scalp from where we are taking our readings. For this thesis, we also computed band powers for channel groups representing major cortical regions:

- **Frontal channels:** Fp1, Fp2, F3, F4, F7, F8

- **Central channels:** Cz, C3, C4
- **Parietal channels:** P3, P4, Pz
- **Occipital channels:** O1, O2
- **Temporal channels:** T7, T8
- **Frontal-left channels:** Fp1, F3, F7
- **Frontal-right channels:** Fp2, F4, F8

For each region r and frequency band b , the ROI band power is defined by

$$P_{r,b} = \frac{1}{|C_r|} \sum_{c \in C_r} P_{c,b} \quad (4.13)$$

where $P_{c,b}$ is power band for channel c and C_r is the set of channels located in region r . Using ROI bands features, we obtain regional power indicators, such as frontal-alpha and parietal-theta power, which play a supplementary role in distinguishing between the attention states.

4.4.5 Comprehensive Spectral Feature Vector:

Since the seven-dimensional target vector (Equation 4.12) captures the neurophysiologically comprehensible attention signatures, we extracted a set of spectral features that served as input for predictive modeling.

The complete feature vector comprises:

- **Global band features:** $P_\delta, P_\theta, P_\alpha, P_\beta, P_\gamma$ (Equation 4.9)
- **Relative band features:** $P_{\delta,\text{rel}}, P_{\theta,\text{rel}}, P_{\alpha,\text{rel}}, P_{\beta,\text{rel}}, P_{\gamma,\text{rel}}$ (Equation 4.12)
- **Attention relevant ratios:** Alpha-beta (α/β) ratio and theta-beta (θ/β) ratio (Equation 4.11)
- **Region-of-Interest (ROI) band powers:** From each of the seven cortical regions and five frequency bands, we calculated regional relative powers using Equation 4.13, which resulted in 35 ROI-features capturing the spatial distribution of spectral activity.

- **Hemispheric asymmetry:** We computed frontal alpha asymmetry using the equation:

$$A_{\text{frontal},\alpha} = \frac{P_{\text{right},\alpha} - P_{\text{left},\alpha}}{P_{\text{right},\alpha} + P_{\text{left},\alpha}} \quad (4.14)$$

where a positive value indicates a right-dominant alpha activity, whereas a negative value indicates left-dominant alpha activity. These frontal alpha asymmetry patterns are linked with an individual's response to varying circumstances [72]

- **Backward compatible feature aliases:** To maintain consistency with EEG analysis conventions, we lastly included alternative representations of global band features using standard naming patterns (e.g., spectral alpha rel mean as an alias for $P_{\alpha,rel}$). These redundant features allowed the predictor model to learn robust representations invariant to naming conventions.

The resulting vector from the above, then had 59 features(dimensions), capturing spectral energy distribution across frequency bands, spatial regions, and cerebral hemispheres. A visual depiction of the feature vector is given in figure 4.2

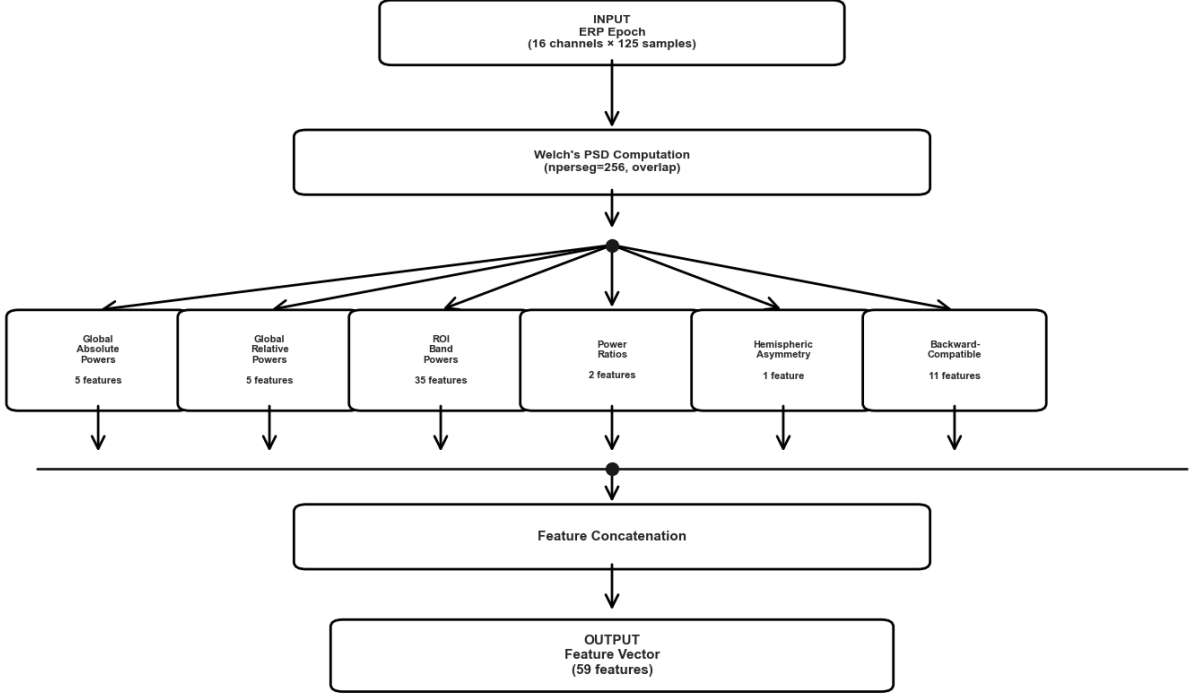


Figure 4.2: Input feature vector for spectral specialist: For input feature vector, each baseline-corrected ERP epoch undergoes Welch’s power spectral density (PSD) computation to estimate frequency-domain characteristics across 0.5-40 Hz. The PSD is then decomposed into six feature categories totaling 59 features, which are then concatenated into a single 59-dimensional vector per epoch, serving as input to the LightGBM spectral specialist.

This output feature vector, containing detailed spectral information, then served as input for the spectral specialist (LightGBM), which then learned to construct neural signatures as given in equation 4.12.

4.5 Attention States Discovery:

From the EEG literature, we have detailed descriptions and interpretations of attention states across frequency bands (section 2.2). However, well-defined thresholds for labeling attention states do not exist in the literature. To address this issue, we used the k-means clustering defined in (section 3.1.1), which allowed us to create a labeling scheme for attention states.

For determining the optimal number of clusters, we first evaluated cluster quality metrics (Calinski-Harabasz Index, Silhouette Index) for k -clusters ranging from 2 to 5. Three

clusters yielded significant separation ($\text{CHI} = 122.89$), corresponding to three distinct attention states: baseline attention, low arousal or drowsiness, and high attention and engagement. We set $k = 3$. The seven-dimensional feature vector $\mathbf{y}_{\text{true}}$ (equation 4.12) served as input for the k-means clustering algorithm. Importantly, we used true band power values computed directly using PSD estimation (section 4.4.2), ensuring that the clustering algorithm learns the true brain activity profiles.

As beta band power $P_{\beta_{\text{rel}}}$ and the alpha-beta ratio ($R_{\alpha_{\text{rel}}/\beta_{\text{rel}}}$) are among the most informative neural signatures for attention modeling [29], we applied feature weighting to emphasize these patterns. Specifically, we assigned higher weights $\mathbf{w} = 3$ to both $P_{\beta_{\text{rel}}}$ and alpha-beta ratio ($R_{\alpha_{\text{rel}}/\beta_{\text{rel}}}$), while all other features retained weights of $\mathbf{w} = 1$. With this weighing strategy, we ensured that the clustering prioritizes attention-relevant neural signatures more, while also preserving the significance of the remaining features during cluster assignment.

After clustering, we ordered the three discovered clusters based on their alpha-beta ratio ($R_{\alpha_{\text{rel}}/\beta_{\text{rel}}}$) from high to low. This ordering was based on our learning from neuroscience literature that a high alpha-beta ratio is an indicator of reduced attention and relaxation. In contrast, a low alpha-beta ratio indicates focused attention and active cognitive engagement in the task at hand [9].

The following labels were then assigned to the ordered clusters based on their band power profiles and attention-relevant signatures.

1. Baseline Attention ($R_{\alpha/\beta} = 1.91$)
2. Low Arousal ($R_{\alpha/\beta} = 1.06$)
3. High Attention ($R_{\alpha/\beta} = 0.52$)

The resulting attention states (clusters) then became interpretable because we assigned labels to these attention states (clusters) using established EEG literature [20, 59].

After clustering, we applied clustering quality metrics to assess clusters' compactness and separability. Specifically, we used the Calinski-Harabasz index and the Silhouette coefficient (sections 3.1.2).

4.6 Architecture of Framework:

The predictive framework for this thesis consists of three sequential stages:

1. In the first stage, we used a multi-output regression model that learns to predict the neurophysiological attention signatures (band powers and ratios) from spectral features (section 4.4.5) extracted from EEG signals.
2. In the second stage, we utilized three meta learning models that integrate specialist predictions along with contextual information to classify each epoch into the attention states discovered through clustering (section 4.5).
3. In the third stage, we quantified uncertainty in each meta-learner’s predictions using model-specific uncertainty estimation methods.

4.6.1 Spectral Specialist:

While we computed true band power values directly from the EEG signal using PSD estimation (section 4.4.2), the spectral specialist learnt to predict these neural signatures from the spectral feature vector. The motivation behind this approach is that the specialist will learn more precise spectral-to-neurophysiological mappings that will generalize well across subjects and paradigms.

For the spectral specialist, we implemented a multi-output regression model that predicts the seven-dimensional target vector (equation 4.12) from the comprehensive feature vector (section 4.4.5) The output vector from the spectral specialist is given as:

$$\hat{\mathbf{y}}_{\text{pred}} = [\hat{P}_{\delta, \text{rel}}, \hat{P}_{\theta, \text{rel}}, \hat{P}_{\alpha, \text{rel}}, \hat{P}_{\beta, \text{rel}}, \hat{P}_{\gamma, \text{rel}}, \hat{R}_{\alpha/\beta}, \hat{R}_{\theta/\beta}] \quad (4.15)$$

Architecture:

We employed Light Gradient Boosting Machine, a gradient boosting model, which is capable of handling tabular data while simultaneously modeling complex, non-linear dependencies among the input features. The specialist model has the following configurations:

- Number of boosting iterations = 200
- Maximum depth of tree = 5

- Learning rate = 0.05
- Minimum samples in each leaf node = 20

We selected these hyperparameters using grid-search tuning, a method for selecting the best set of hyperparameters to optimize the model’s performance and to have a good balance between bias and variance, while preserving the model’s capacity to capture underlying spectral-to-neurophysiological relationships.

We implemented a multi-output strategy by training seven separate LightGBM regressors (one per target) for each of the seven targets (using scikit-learn’s MultioutputRegressor wrapper), all of which shared the same input features.

4.6.2 Meta-Learning Framework:

In the meta-learning stage, we performed multiclass classification of attention states discovered through clustering (section 4.5). Unlike the specialist (which predicts continuous band powers), the meta-learner categorizes each epoch into one of three distinct attention states: Baseline attention, Low arousal, and High attention.

As an input to the meta-learner, we combined seven specialist predictions (five relative band powers and two band ratios from equation 4.15) with nine contextual features encoded as a one-hot vector. The contextual features represent experimental paradigms (oddball vs video), alongside their specific condition (target, non-target, car, animal, surprise). These contextual features enabled the meta-learner to learn paradigm-specific attention states. The resulting input vector (meta-feature vector) then had 16 features per epoch.

We then evaluated three meta learners to determine which best integrates specialist predictions and contextual information for attention state classification.

1. **Logistic Regression:** A linear classifier that learns a weighted combination of input features to estimate class probabilities for different attention states. We configured it with balanced class weights to address class imbalance and a one-vs-rest multiclass strategy.
2. **XGBoost Classifier:** A tree-based ensemble technique that outputs non-linear decision boundaries. Using this classifier, we modeled complex interactions between specialist predictions and context. We configured the XGBoost classifier

with the following hyperparameters obtained from a grid search hyperparameter tuning strategy.

- Total number of boosting trees = 250
- Maximum depth of each tree = 4
- Learning rate = 0.06
- L2 regularization (λ) = 1

3. **Transformer-based Meta-Learner:** A self-attention architecture that learns to weight each meta-feature based on its relevance, allowing the model to reweigh each feature’s contribution on a per-sample basis adaptively. In this thesis, we configured the transformer as follows:

- **Embedding layer:** A single layer that projects a 16-dimensional input to a 64-dimensional hidden representation.
- **Transformer Encoder:** Two stacked encoder layers with four attention heads and a 128-dimensional feedforward network with the dropout rate set to 0.1.
- **Classification head:** A linear layer mapping hidden representation to three discovered attention states. We trained the transformer for 50 epochs using the Adam optimizer (learning rate = 0.01), with a batch size of 32 and a cross-entropy loss function.

4.6.3 Uncertainty in Meta learner’s Predictions:

After obtaining the attention state predictions from each of the three meta learners, we then quantified the uncertainty associated with each model’s prediction. For all three meta learners, we estimated prediction uncertainty by analyzing each model’s confidence in its class prediction. This uncertainty is quantified as Shannon entropy [36] over the predicted probability distribution:

$$H(p) = - \sum_i p_i \log_2(p_i) \quad (4.16)$$

where p_i is the predicted class probability for attention states $i \in 0, 1, 2$. Higher entropy indicates greater prediction uncertainty, with a maximum value of 1.58, representing

uniform uncertainty (equal probability across all three states), while lower entropy indicates confident predictions.

For Logistic regression, we computed uncertainty from calibrated probability distributions. We first calibrated the model’s output probabilities using Platt scaling (section 3.5.1). This step adjusts the probability outputs to align with actual classification accuracy. We then computed predictive entropy from these calibrated probabilities.

For the XGBoost classifier, we leveraged the ensemble nature of gradient boosting to estimate uncertainty. Since the classifier is aggregating predictions from 250 boosting trees to produce class probability estimates, we computed predictive entropy from these ensemble-aggregated probabilities. High entropy reflects uncertainty in the ensemble-aggregated class probability distribution, indicating low confidence in the predicted class.

For the Transformer, we utilized MC dropout (section 3.5.2) by randomly deactivating 10% neurons (dropout rate = 0.1) in the network to zero during network training. During testing, we kept dropout active and performed 50 forward passes with different random dropout neurons to induce stochasticity in the network. This resulted in a distribution of predictions $p_1, p_2, \dots, p_{50}$. We then computed the mean prediction $\bar{p} = \frac{1}{N} \sum_{t=1}^{50} p_t$ and calculated predictive entropy $H(\bar{p})$ from this averaged distribution.

After using model-specific uncertainty quantification methods, we employed the Mann-Whitney U test to verify that uncertainty estimates systematically distinguish correct from incorrect predictions. This non-parametric test is particularly suitable for our sample size, as it makes no assumptions about distributional normality and remains statistically powerful even with small error counts. For this test, if the p -value for uncertainty difference is less than the predetermined significance level ($\alpha = 0.05$), we reject the null hypothesis and conclude that incorrect predictions have significantly higher uncertainty than correct predictions, indicating good calibration.

4.7 Cross Validation Strategy:

To evaluate trained machine learning and deep learning models’ performance on new subjects, we used a Leave-One-Subject-Out (LOSO) cross-validation strategy. In LOSO, we held out one subject as the test subject and trained the models on the remaining sub-

jects, repeating this process so each subject served as the test subject exactly once. We then calculated the mean accuracy and mean error to quantify each model’s performance.

The motivation behind using the LOSO strategy is that the EEG studies are experimental subject-based; hence, we can’t simply split the total experimental data into an 80-20 train-test data split strategy. This would introduce bias in the model performance, and this would hurt the generalization capabilities of the trained models.

Within each fold, we trained both spectral specialist and meta-learners exclusively on $N-1$ subjects and then tested on the held-out subject, ensuring genuine generalization to new subjects.

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5.1 Event Related Potential (ERP) Analysis:

The goal of ERP analysis was to verify the presence of attention-related neural markers and evaluate event-locked approaches. For this purpose, we first extracted 418 event-locked epochs, spanning a time interval from 200 ms pre-stimulus to 800 ms post-stimulus. These epochs comprised oddball targets (n=76), oddball non-targets (n=103), video car crashes (n=134), video animal (n=84), and video surprise events (n=21). We then applied baseline correction for epoch normalization using the interval from -200 ms to 0 ms before grand averaging.

Before conducting statistical tests for ERP detection, we verified the assumptions of normality (section 3.4.1) and homoscedasticity (section 3.4.1). Although the data exhibited deviations from the strict normality condition, which is often observed in EEG feature distributions with a relatively large sample size (418 epochs), the condition of variance homogeneity was satisfied. Given the relatively large sample size, we proceeded with robust parametric testing, relying on the Central Limit Theorem, which states that for sufficiently large sample size n , the sampling distribution of sample mean $\bar{X}$ converges in distribution to the normal distribution regardless of underlying data distribution [49].

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

5.1.1 P300 Component for Oddball Task:

In figure 5.1, we visualized average ERPs for the oddball task at six electrodes, focusing mainly on the centro-parietal region of the scalp where P300 usually has maximum amplitude. In the figure, the x-axis represents time in seconds, with the y-axis representing the amplitude of the brain wave in microvolts (μV). The vertical dashed red line represents the stimulus onset (at 0 sec). The gray-shaded region depicts the pre-stimulus interval (-200 ms to 0 ms) used for baseline correction period, where all amplitudes were

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corrected by subtracting the mean voltage during this interval to remove pre-existing DC offsets and isolate stimulus-evoked activity. Target trials (red, $n=76$) and non-target trials (blue, $n=103$) represent grand averages computed across all valid epochs per condition. The purple dashed line depicts the difference wave (target minus non-target), which isolates attention-specific neural activity by canceling out shared sensory processing common to both conditions. Finally, the yellow-shaded region represents the P300 window, which spans from 300 ms to 600 ms post-stimulus.

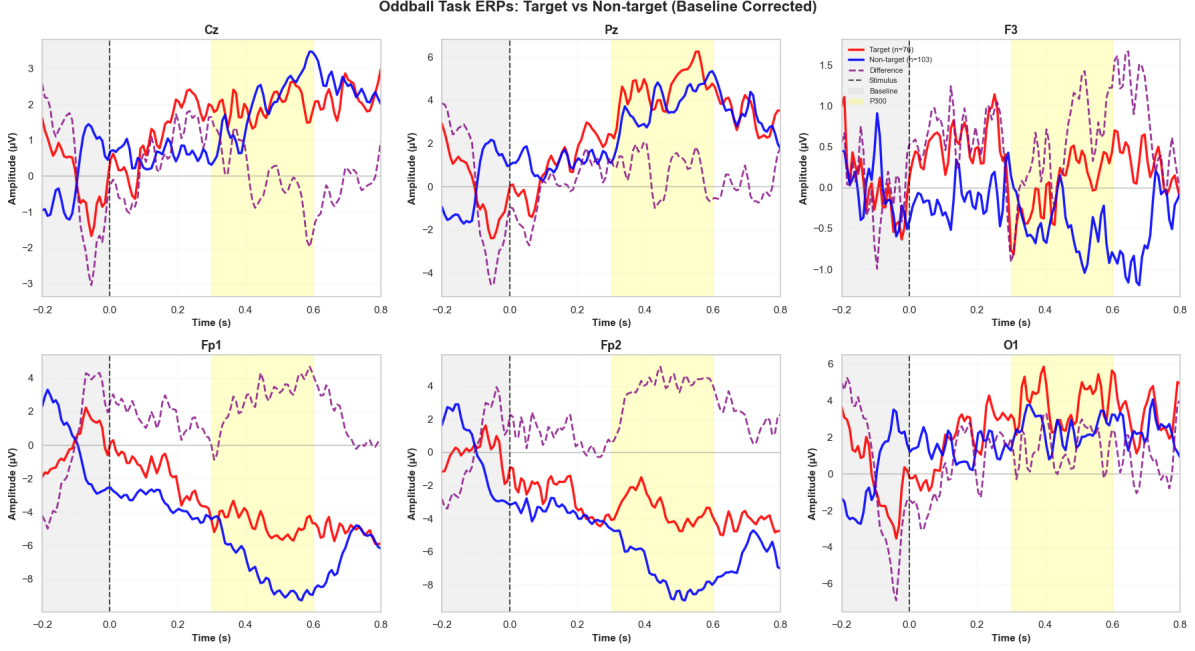


Figure 5.1: Oddball Task Event-Related Potentials (ERPs): Target versus Non-target. Each subplot displays time on the x-axis (spanning -200 ms to +800 ms) and ERP amplitude on the y-axis in microvolt (μV). The vertical dashed red line at 0 seconds marks the occurrence of the stimulus. The gray-shaded region (-200 ms to 0 ms) indicates the pre-stimulus baseline period. Target trials are represented by a red line, and non-target trials are represented by a blue line. The purple dashed line depicts the difference wave. The yellow-shaded region represents the P300 window.

For the statistical analysis, we extracted the mean amplitude during the P300 window for each of the six electrodes for each epoch and conducted an independent sample t-test (from section 3.4.1). The results from the t-test showed that the target ERP component ($n=76$) has a mean value $Mean = 2.57\mu V$ and a standard deviation of $\sigma = 16.68\mu V$, which did not significantly differ from the non-target component ($n=103$), which has a mean value $Mean = 2.52\mu V$ and standard deviation $\sigma = 14.96\mu V$. The mean difference was not

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sufficient ($\Delta M = 0.05 \mu V$, 95% CI : $[-4.68, 4.79] \mu V$), which t-test confirmed (from section 3.4.1), $t(177) = 0.021$, $p = 0.983$. The effect size (from section 3.4.1) was also very small, Cohen's $d = 0.003$ (95% CI : $[-0.29, 0.30]$).

These results indicate that the P300 response did not reach statistical significance in this dataset. This absence may be attributed to several factors, including participant fatigue, the relatively low cognitive demand of the oddball task, inter-subject variability in ERP morphology and latency, and the limited number of participants. Nevertheless, visual inspection of the ERPs reveals qualitative evidence of a P300-like deflection within the 300-600 ms post-stimulus window, suggesting that neural attention-related responses did occur despite failing to reach statistical significance.

5.1.2 Video Paradigm Analysis:

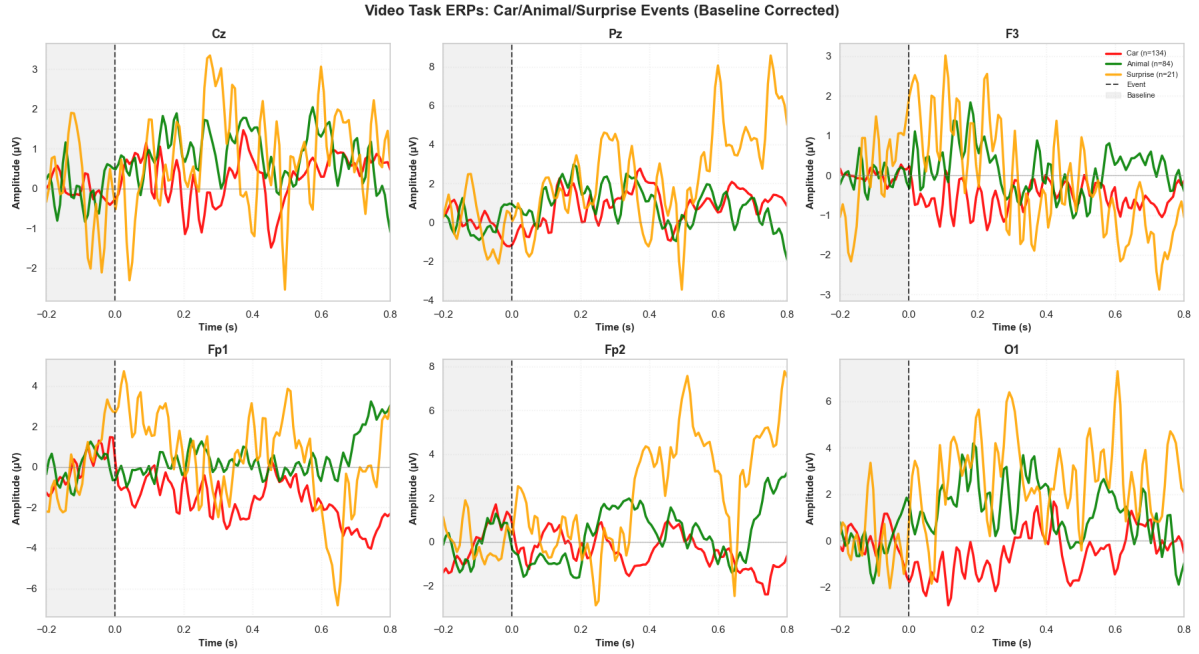


Figure 5.2: Event Related Potential (ERPs) for Naturalistic Video Paradigms: For each subplot, the gray-shaded region (-200 ms to 0 ms) indicates the pre-stimulus baseline period. The red line represents the car crash event, the green line indicates animal presence events, whereas the orange line represents surprise events. The x-axis is the time axis that spans from -200 ms pre-event to +800 ms post-event, with the vertical dashed line marking the event onset at 0 seconds (i.e., the frame at which the video label first appeared). The y-axis depicts ERP amplitude on the y-axis in microvolt μV .

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In figure 5.2, we visualized average ERPs for video events at six electrodes located at the centro-parietal region of the scalp. The car crash event is represented by a red line with a total number of events $n = 134$. The surprise events are represented by an orange line with a total number of events $n = 21$, and the animal presence events are represented by a green line with a total number of events $n = 84$. The event onset is marked by a vertical dashed red line, while the gray-shaded region represents the pre-stimulus baseline interval.

The visual analysis suggested that surprise events evoked the strongest responses, especially at electrodes Pz and Cz, where the amplitude reached $\approx 0.8\mu V$. The car crash segments produced sustained amplitude between $1\mu V$ to $4\mu V$ throughout the ERP window (300 - 600ms), where the max amplitude ($\approx 4\mu V$) was observed at the occipital electrode O1. The animal presence events indicated a consistent amplitude ranging between $-2\mu V$ to $+2\mu V$.

For assessing the statistical differences in mean ERP amplitudes across video paradigms, we conducted one-way ANOVA (from section 3.4.1) on mean amplitudes extracted from the defined ERP window. The ANOVA test resulted in an F-statistic value equal to 0.289 with a non-significant p-value equal to 0.749, indicating no statistically significant differences in mean ERP amplitudes among the three video conditions. This suggested that the naturalistic video events produced relatively consistent ERP patterns across all video paradigms.

5.2 Clustering Outputs:

To classify each epoch into an attention state label, we used clustering (for three identified attention states) as an attention state label allocation strategy, with clustering performed independently within each LOSO fold (section 4.5). From the table 5.1, the baseline attention state contained 30.6% of the data, indicating a moderately attentive and routine processing state, as supported by the theta-dominance ($\theta = 0.295 \pm 0.087$ and $\theta/\beta = 3.00 \pm 2.10$) and a moderate beta value $\beta = 0.125 \pm 0.050$. The low arousal state contained 28.9% of data characterized by a low beta value ($\beta = 0.103 \pm 0.060$) and a high delta band value ($\delta = 0.637 \pm 0.116$), indicating reduced alertness and drowsiness. The high attention state contained 40.4% of the data, representing focused engagement, supported by a relatively high beta value ($\beta = 0.304 \pm 0.079$) and low alpha-beta value

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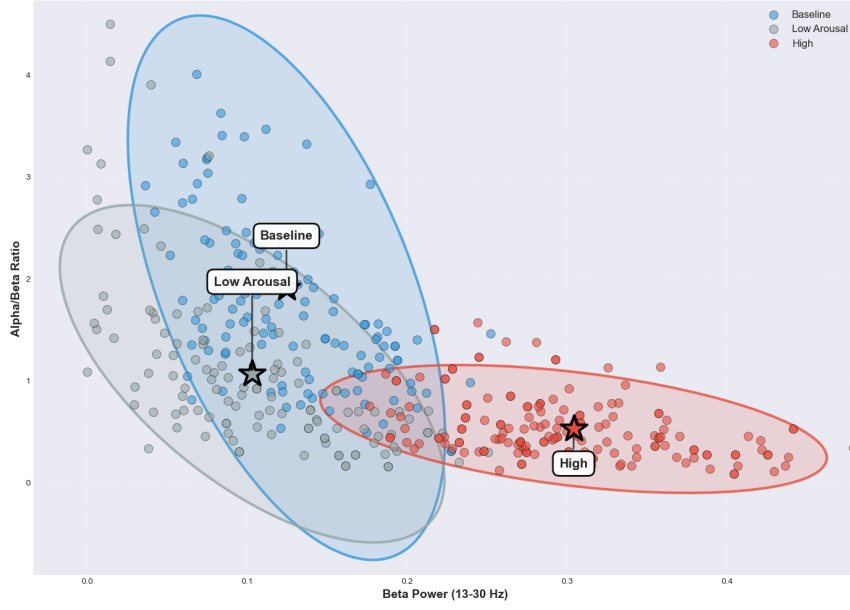
($\alpha/\beta = 0.52 \pm 0.31$). We assigned numeric labels (0, 1, 2) to the clusters in descending order of the alpha–beta ratio, with label 0 corresponding to the baseline attention state (highest alpha-beta ratio), ensuring consistent state definitions across all subjects.

Table 5.1: Neurophysiological profiles of discovered attention states

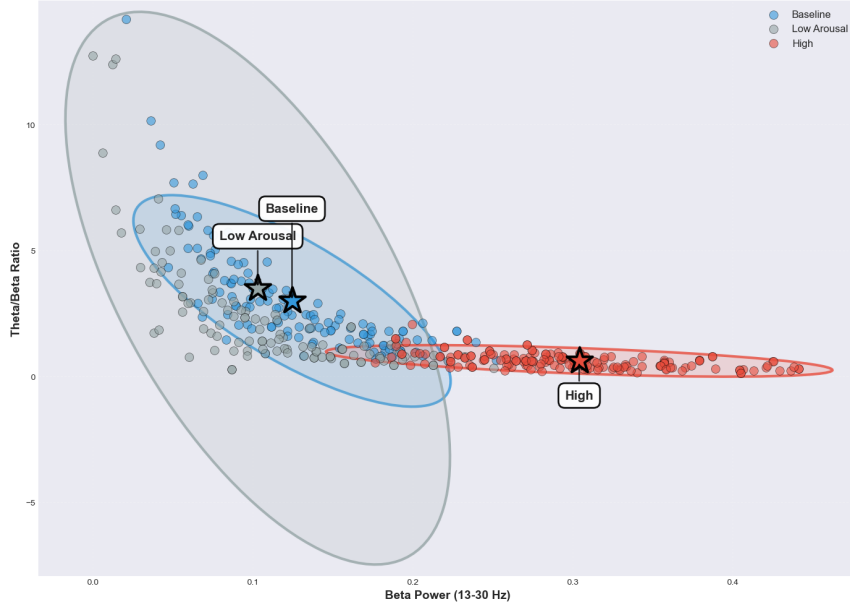
State	N	α/β	θ/β	β	δ	θ	Profile
	(%)		(mean $\pm$ SD, relative power)				
Baseline Attention	128 (30.6)	1.91 ± 1.34	3.00 ± 2.10	0.125 ± 0.050	0.357 ± 0.095	0.295 ± 0.087	θ -dominant, routine
Low Arousal	121 (28.9)	1.06 ± 0.83	3.50 ± 5.49	0.103 ± 0.060	0.637 ± 0.116	0.161 ± 0.069	δ -dominant, drowsy
High Attention	169 (40.4)	0.52 ± 0.31	0.61 ± 0.31	0.304 ± 0.079	0.312 ± 0.105	0.173 ± 0.069	β -dominant, focused

From the cluster evaluation metrics, we obtained a Calinski-Harabasz index of 122.89 and a Silhouette coefficient value of 0.266, indicating optimal cluster separation with meaningful inter-cluster variance. Figure 5.3 visualizes the clustering structure in beta power versus alpha–beta and theta–beta feature spaces.

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(a) Alpha-Beta ratio vs Beta band clusters



(b) Theta-Beta ratio vs Beta band cluster

Figure 5.3: 2D projection of clusters corresponding to distinct attention states: The blue cluster represents the baseline attention state, the grey cluster corresponds to the low arousal state, and the red cluster indicates the high attention state. Clusters corresponding to the three attention states are ordered in descending order based on the alpha-beta ratio.

5.3 Spectral Specialist Performance Analysis:

We trained a Light Gradient Boosting Machine (from section 3.2.1) as a spectral specialist to predict seven neural attention markers from input spectral features using a leave-one-subject-out (LOSO) cross-validation strategy across 9 subjects (from section 4.6.1).

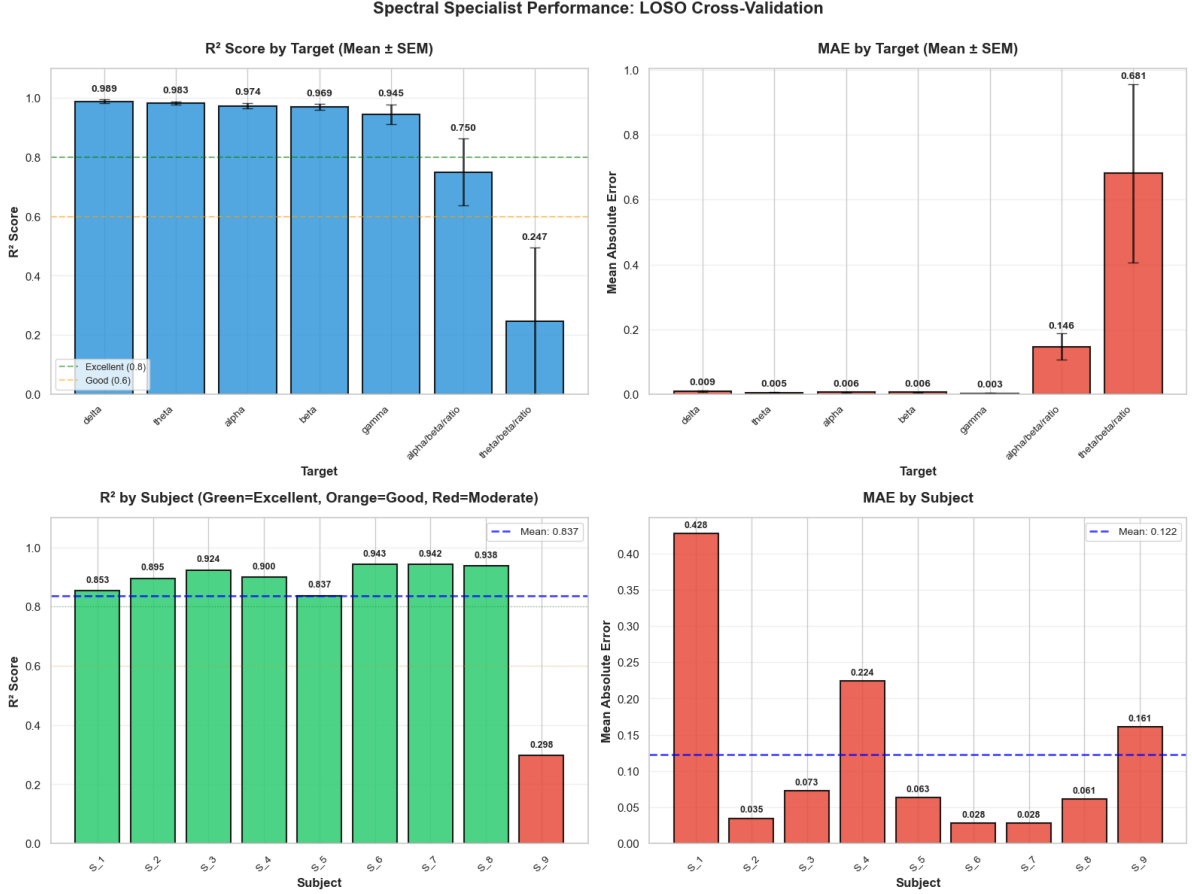


Figure 5.4: Spectral Specialist Performance: LOSO Cross-Validation Results: **Top row:** Average performance across 9 subjects for seven prediction targets. **(Left)** R^2 scores showing excellent performance ($R^2 > 0.94$) for five band powers (delta, theta, alpha, beta, gamma), good performance for alpha-beta ratio ($R^2 = 0.750$), and moderate performance for theta-beta ratio ($R^2 = 0.247$). **(Right)** Mean absolute error (MAE) confirming high accuracy for band powers (MAE < 0.01) and higher variability for theta-beta ratio (MAE = 0.681). **Bottom row:** Per-subject performance. **(Left)** Eight subjects achieved $R^2 > 0.8$. The dashed line indicates the overall mean $R^2 = 0.837$. **(Right)** Per-subject MAE with overall mean MAE = 0.122 (dashed line). Subject S_1 shows the highest error (MAE = 0.428), while the remaining subjects maintain MAE < 0.25 . Error bars represent the standard error of the mean (SEM) across LOSO folds.

The figure 5.4 shows a detailed overview of performance metrics across all 9 subjects. The spectral specialist achieved excellent overall performance for predicting five EEG band powers with a mean $R^2 > 0.90$ and $MAE < 0.01$, indicating that the specialist can reproduce five power bands with high accuracy. The alpha-beta ratio showed good performance with $R^2 = 0.750$ and $MAE = 0.146$, whereas the theta-beta ratio achieved $R^2 = 0.247$ and $MAE = 0.681$. The low R^2 (0.247) and high MAE (0.681) indicated instability in the theta-beta ratio, likely due to its sensitivity to small beta-band values in the denominator and increased inter-trial variance.

These performance metrics are also reflected in per-subject analysis, where eight out of 9 subjects have $R^2 > 0.83$ with an overall $MAE = 0.122$. These results validated the robustness of the spectral specialist across subjects and confirmed its ability to capture underlying data distribution.

5.4 Meta-Learner Performance:

During the meta-classification stage, we evaluated three architectures for classifying each epoch into discovered attention states. We implemented three models: a multi-class logistic regression as a baseline model, XGBoost, and a custom transformer with attention-based fusion of spectral and contextual features. The clustering labels (from section 5.2) served as training target labels, converting unsupervised attention states discovery into a supervised classification task.

All three models received a 16-dimensional input vector combining seven spectral predictions from LightGBM (5 bands, 2 ratios) with nine one-hot encoded contextual features indicating experimental paradigm (oddball vs. video) and condition (target, non-target, car, animal, surprise). This context-aware representation enabled meta-learners to account for paradigm-specific variations in attention signatures. Each meta learner predicted a single numeric value (0,1, or 2) representing the predicted attention states. These numeric labels were then mapped to the neurophysiological states defined in section 5.2.

All three meta-learners achieved an excellent predictive performance (figure 5.5), indicating strong consistency between the discovered attention states and the learned decision boundaries.



Figure 5.5: Meta-learners’ performance comparison across three architectures: All models achieved accuracy >95% with minimal discrepancy between accuracy and balanced accuracy, indicating consistent performance across all three attention states without majority class bias.

Logistic regression achieved the highest accuracy among the three, with an accuracy of 97.4% and an F1 score of 0.972. XGBoost achieved 96.4% accuracy with an F1 score of 0.963. Transformer achieved 95.7% accuracy with an F1 score equal to 0.956. A highly similar value of accuracy and balanced accuracy for each of the three learners indicated consistent classification quality across all three attention states.

The minimal discrepancy between accuracy and balanced accuracy for all models confirmed consistent classification quality across the three attention states, indicating that performance is not driven by majority class bias. The predictive performance of the logistic regression model is shown in figure 5.6, showing that misclassifications occur primarily between Baseline Attention and Low Arousal states, revealing the transitional nature of theta-dominant epochs between routine processing and drowsiness.

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Subject	Paradigm	Condition	Attention State Predictions: All Conditions		Beta	Delta	$\alpha\beta$	$\square$
			True State	Predicted State				
S_1	Oddball	Oddball Nontarget	Low Arousal	Low Arousal	0.114	0.480	0.71	$\square$
S_1	Oddball	Oddball Nontarget	Baseline Attention	Baseline Attention	0.060	0.371	3.13	$\square$
S_1	Oddball	Oddball Nontarget	Baseline Attention	Baseline Attention	0.055	0.384	3.34	$\square$
S_1	Oddball	Oddball Target	Baseline Attention	Baseline Attention	0.086	0.463	1.26	$\square$
S_1	Oddball	Oddball Target	Baseline Attention	Baseline Attention	0.124	0.450	0.74	$\square$
S_1	Oddball	Oddball Target	Baseline Attention	Baseline Attention	0.087	0.476	1.28	$\square$
S_3	Video Animal	Video Animal	Baseline Attention	Low Arousal	0.132	0.484	1.94	$\square$
S_3	Video Animal	Video Animal	Baseline Attention	Baseline Attention	0.172	0.344	1.04	$\square$
S_3	Video Animal	Video Animal	Baseline Attention	Low Arousal	0.108	0.507	2.29	$\square$
S_1	Video Car	Video Car	Low Arousal	Low Arousal	0.080	0.991	3.27	$\square$
S_1	Video Car	Video Car	Low Arousal	Low Arousal	0.061	0.688	0.93	$\square$
S_1	Video Car	Video Car	Low Arousal	Low Arousal	0.004	0.896	1.56	$\square$
S_3	Video Surprise	Video Surprise	Baseline Attention	Baseline Attention	0.154	0.363	1.81	$\square$
S_3	Video Surprise	Video Surprise	High Attention	High Attention	0.224	0.480	0.38	$\square$
S_3	Video Surprise	Video Surprise	Baseline Attention	Baseline Attention	0.109	0.235	1.36	$\square$

Figure 5.6: Epoch-Level Predictions from Logistic Regression Meta-Learner: Each row represents a single epoch, including its subject identifier, paradigm, condition, ground-truth attention state (from clustering), predicted state, and key band-power features (beta, delta, alpha-beta ratio).

The superior performance of logistic regression compared to more advanced learners indicated that the discovered attention states are approximately linearly separable in the 16-dimensional meta-feature space. This linear separability also confirmed the clustering quality, as well-separated clusters exhibit minimal overlap and linear decision boundaries.

The high performance across all architectures (accuracy $> 95\%$) validated both the neurophysiological meaningfulness of discovered attention states and the quality of spectral predictions from LightGBM.

5.5 Uncertainty Analysis:

After attention state classification, we then quantified uncertainty for all three meta-learners using model-specific methods.

For logistic regression, we used calibrated predictive entropy. For XGBoost, we used an ensemble-based entropy approach, and for the transformer, we employed MC dropout. The table 5.2 summarizes the uncertainty statistics across all 418 epochs for each meta learner. We calculated uncertainty using Shannon entropy, resulting in uncertainty

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values ranging from 0 (complete certainty) to $\log(3) \approx 1.099$ (complete uncertainty).

Table 5.2: Mean Uncertainty Value for All Three Meta Models

Model	Mean $\pm$ Std
Logistic Regression	0.3477 ± 0.2721
XGBoost	0.1398 ± 0.2248
Transformer	0.0581 ± 0.1446

The Transformer achieved the lowest mean uncertainty (0.0581), indicating high confidence in its predictions. XGBoost exhibited a moderate mean uncertainty (0.1398). In contrast, Logistic Regression showed the highest mean uncertainty (0.3477) due to its probabilistic calibration via Platt scaling, which deliberately increases uncertainty for ambiguous cases to improve reliability.

The table 5.3 presents mean uncertainty values for correct versus incorrect predictions, along with statistical validation using the Mann-Whitney U test (one-sided, alternative: incorrect $>$ correct; good calibration).

Table 5.3: Mean predictive uncertainty associated with correct and incorrect predictions. Values are reported as mean $\pm$ standard deviation.

Model	Correct	Incorrect	Difference	p -value
Logistic Regression	0.3361 ± 0.2654 ($n = 407$ correct, $n = 11$ incorrect)	0.7769 ± 0.1343	+0.4408	1.27×10^{-6}
XGBoost	0.1241 ± 0.2115 ($n = 401$ correct, $n = 17$ incorrect)	0.5117 ± 0.2095	+0.3877	7.43×10^{-9}
Transformer	0.0456 ± 0.1243 ($n = 402$ correct, $n = 16$ incorrect)	0.3718 ± 0.2356	+0.3262	6.66×10^{-8}

All three models demonstrated strong separation between correct and incorrect predictions (all $p < 10^{-6}$), with incorrect predictions exhibiting $2\times$ higher uncertainty than correct predictions (measured via Cohen’s d effect sizes: LR: d=2.096, XGBoost: d=1.842, Transformer: d=1.732; all exceeding the "large effect" threshold of 0.8). XGBoost achieved the highest statistical significance ($p = 7.43 \times 10^{-9}$) for uncertainty associated with correct and incorrect predictions, indicating that the ensemble-based entropy approach successfully captured epistemic uncertainty in the model’s predictions. The

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Transformer showed a difference of $+0.3262$ between uncertainty associated with correct and incorrect prediction (with $p = 6.66 \times 10^{-8}$), while Logistic Regression achieved the largest difference $+0.4408$ (with $p = 1.27 \times 10^{-6}$). These results confirmed that all three models reliably distinguish between confident and uncertain predictions, which is a critical requirement for model deployment in safety-critical applications.

5.5.1 Uncertainty Across Paradigms:

Next, we analyzed uncertainty patterns across different experimental paradigms to identify the contexts in which attention states were most ambiguous. Figure 5.7 represents mean predictive uncertainty per paradigm for each meta learner.

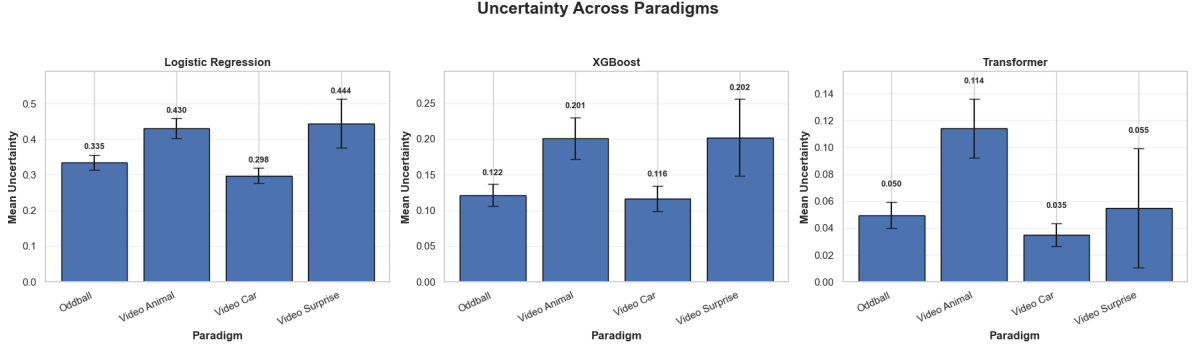


Figure 5.7: Mean predictive uncertainty across experimental paradigms. Error bars here are an indicator of the standard error of the mean and represent ± 1 standard deviation.

For Logistic Regression, video paradigms produced higher uncertainty than the oddball task, with video surprise showing the highest mean uncertainty (0.4440), followed by video animal (0.4305), video car (0.2977), and oddball (0.3349). XGBoost and Transformer showed lower uncertainty overall, with reversed paradigm sensitivity: Oddball tasks produced slightly elevated uncertainty (XGBoost: 0.122, Transformer: 0.05) compared to video paradigms (video car had the lowest uncertainty for both models: XGBoost 0.116, Transformer 0.035). These paradigm-dependent patterns reflected differences in how models leverage contextual features: Logistic Regression relies primarily on spectral predictions (which exhibit higher variance in naturalistic video paradigms), while XGBoost and Transformer more effectively exploit paradigm-specific signatures encoded in one-hot contextual inputs, reducing uncertainty for video conditions. Across all three models, paradigm-related differences were moderate (10–33%), indicating that

meta-learners generalized well across experimental contexts.

5.5.2 Uncertainty Across Attention States:

We examined uncertainty distributions for each of the three discovered attention states to identify which states were most difficult to classify. Figure 5.8 represents mean uncertainty by attention state.

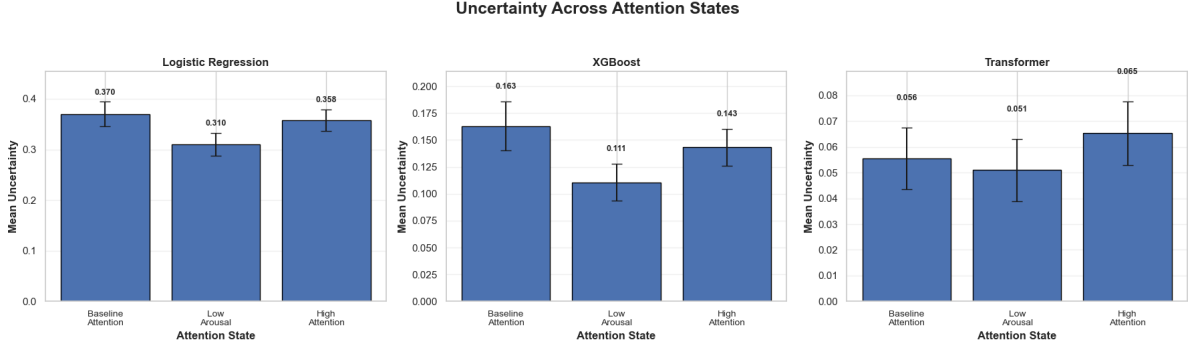


Figure 5.8: Mean predictive uncertainty across each attention state. Error bars here are an indicator of the standard error of the mean and represent ± 1 standard deviation.

For all three models, the Low Arousal state (state 1, δ -dominant) exhibited the lowest uncertainty (LR: 0.310, XGBoost: 0.111, Transformer: 0.051), reflecting the highly distinctive spectral signature of delta-dominant drowsiness with suppressed beta activity. Baseline Attention (state 0, θ -dominant) showed the highest uncertainty for Logistic Regression and XGBoost (LR: 0.37, XGBoost: 0.1630), while High Attention (state 2, β -dominant) produced moderate uncertainty values (LR: 0.358, XGBoost: 0.143, Transformer: 0.065).

The elevated uncertainty for Baseline Attention likely reflects the transitional nature of theta-dominant epochs, which can represent either routine processing (moderate engagement) or the onset of drowsiness, creating ambiguity in the spectral profile. In contrast, Low Arousal’s low uncertainty confirms that delta-dominant patterns are unambiguous markers of reduced vigilance, while High Attention’s beta-dominant signature may exhibit some overlap with Baseline Attention during fluctuations in engagement intensity.

6 Discussion

In this thesis, we developed a meta-learning framework for robust prediction of attention states from raw EEG signals, addressing three key limitations in current BCI research: single-paradigm limitations, circular labeling of attention states, and the lack of predictive uncertainty estimation (section 2.6). Using multi-paradigms EEG recordings from nine subjects, we discovered three distinct attention states through unsupervised clustering (section 4.5) and achieved a classification accuracy of 97.4% (section 5.4) while explicitly quantifying predictive uncertainty.

The proposed data-driven approach for discovering attention states represents an improvement over traditional predefined labeling methods. The discovery of three distinct states via clustering supports the view that attention states may reflect actual neurophysiological patterns rather than arbitrary statistical partitions. In addition, the cross-paradigm training strategy verified that trained models generalize well across different experimental contexts.

The uncertainty quantification analysis (section 5.5) further revealed that all three meta-learners can reliably distinguish between correct and incorrect predictions, with incorrect predictions having two to four times higher uncertainty than the correct predictions (all $p < 10^{-6}$). This indicates that models are capable of recognizing the situations when they lack confidence in predictions and points towards good calibration.

While this thesis demonstrates promising results, there are several opportunities for future expansion. First, while the study was conducted with nine participants, expanding the dataset to include a larger number of subjects (e.g., 20–40 participants) would further strengthen statistical power and enable more robust statistical validation, as well as improved characterization of inter-subject variability in attention patterns. Second, although four experimental paradigms (oddball, car crash, animal video, video surprise) were sufficient to demonstrate cross-paradigm robustness, incorporating a wider range of experimental conditions and real-world, high-stress scenarios would further validate the

generalization capability of the proposed framework. Finally, while all experiments were performed in a controlled environment to ensure methodological rigor, future real-world deployments will play a crucial role in evaluating long-term stability, robustness, and practical usability of the proposed framework.

Another promising direction for future work is the inclusion of generative AI models to enhance both data augmentation and data variability. Conditional Variational Auto-Encoders (CVAEs) and Generative Adversarial Networks (GANs) could help in synthesizing a realistic EEG epoch that corresponds to a specific attention state. Furthermore, this will also solve the issue of data scarcity, which is a persistent challenge in BCI research. Finally, deploying this framework in real-world safety-critical applications will help in assessing the practical utility of the proposed pipeline while revealing the limitations in the current pipeline that are not observable in a controlled environment.

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Declaration of Originality

I hereby declare that this thesis document represents my original work and has been written by me. All sources of information, including published and unpublished work is clearly indicated and properly cited in accordance with academic conventions.

During the preparation of this thesis, I used ChatGPT and other language assistance tools to enhance the readability of this thesis through grammar checks, sentence restructuring, and refined phrasing, while preserving my original ideas and findings. After using a specific tool/service, I reviewed and edited the content as needed and take full responsibility for the content of this document.

Ahmed Sameed
February 8, 2026